```
clc;
clear;
close all;

%% IIR FILTER COEFFICIENTS
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1, N-1)];  % Unit impulse
n = 0:N-1;

%% 1. DIRECT FORM-I AND DIRECT FORM-II
y1 = filter(b, a, x);   % Direct Form-I
y2 = filter(b, a, x);   % Direct Form-II equivalent

%% 2. CASCADE REALIZATION
[sos, g] = tf2sos(b, a);

yc = g*x;

for k = 1:size(sos, 1)
    yc = filter(sos(k, 1:3), sos(k, 4:6), yc);
end

%% 3. PARALLEL REALIZATION
[r, p, k] = residuez(b, a);

yp = filter(k, 1, x);

for i = 1:length(r)
    yp = yp + filter(r(i), [1 -p(i)], x);
end

%% 4. DISPLAY COEFFICIENTS
disp('Numerator coefficients b =');
```

```
Numerator coefficients b =
```

```
disp(b);
```

```
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
disp('Denominator coefficients a =');
```

```
Denominator coefficients a =
```

```
disp(a);
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
disp('Second Order Sections (Cascade) =');
```

```
Second Order Sections (Cascade) =
```

```
disp(sos);
```

```
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327
```

```
disp('Cascade Gain =');
```

```
Cascade Gain =
```

```
disp(g);
```

```
    0.0048
```

```
disp('Residues (Parallel) =');
```

```
Residues (Parallel) =
```

```
disp(r);
```

```
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i
```

```
disp('Poles (Parallel) =');
```

```
Poles (Parallel) =
```

```
disp(p);
```

```
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i
```

```
disp('Direct term =');
```

```
Direct term =
```

```
disp(k);
```

```
    0.0256
```

```
%% 5. ERROR AND PERCENTAGE DEVIATION
E = [max(abs(y1-y2));
     max(abs(y1-yc));
     max(abs(y1-yp))];
```

```
reference = max(abs(y1));

if reference > 0
    D = E/reference*100;
else
    D = zeros(size(E));
end

Structure = ["DF-I"; "DF-II"; "Cascade"; "Parallel"];
Maximum_Error = [0; E];
Deviation_percent = [0; D];

T = table(Structure, Maximum_Error, Deviation_percent);

disp('Comparison Table:');
```

```
Comparison Table:
```

```
disp(T);
```

Structure	Maximum_Error	Deviation_percent
"DF-I"	0	0
"DF-II"	0	0
"Cascade"	1.9221e-15	8.242e-13
"Parallel"	1.9811e-15	8.4949e-13

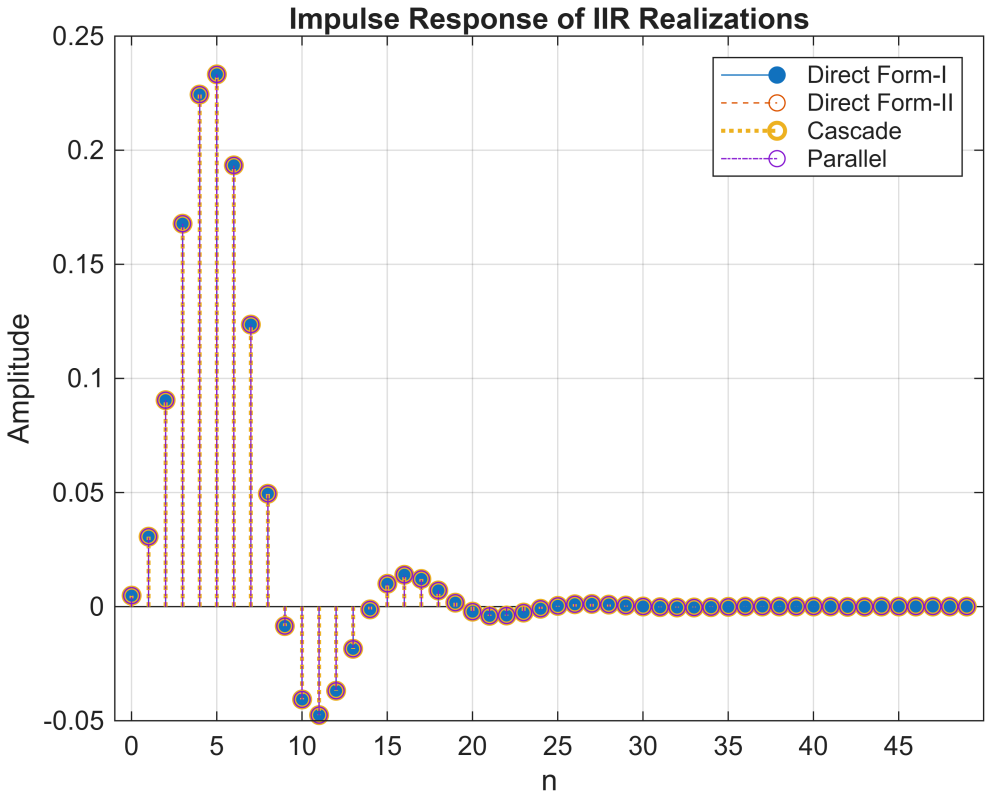
```
%% 6. IMPULSE RESPONSE COMPARISON
figure;

stem(n, y1, 'filled');
hold on;
stem(n, y2, '--');
stem(n, yc, ':', 'LineWidth', 1.5);
stem(n, yp, '-.');
```

```
Warning: Using only the real component of complex data.
```

```
grid on;
xlabel('n');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');

legend('Direct Form-I', 'Direct Form-II', ...
       'Cascade', 'Parallel');
```

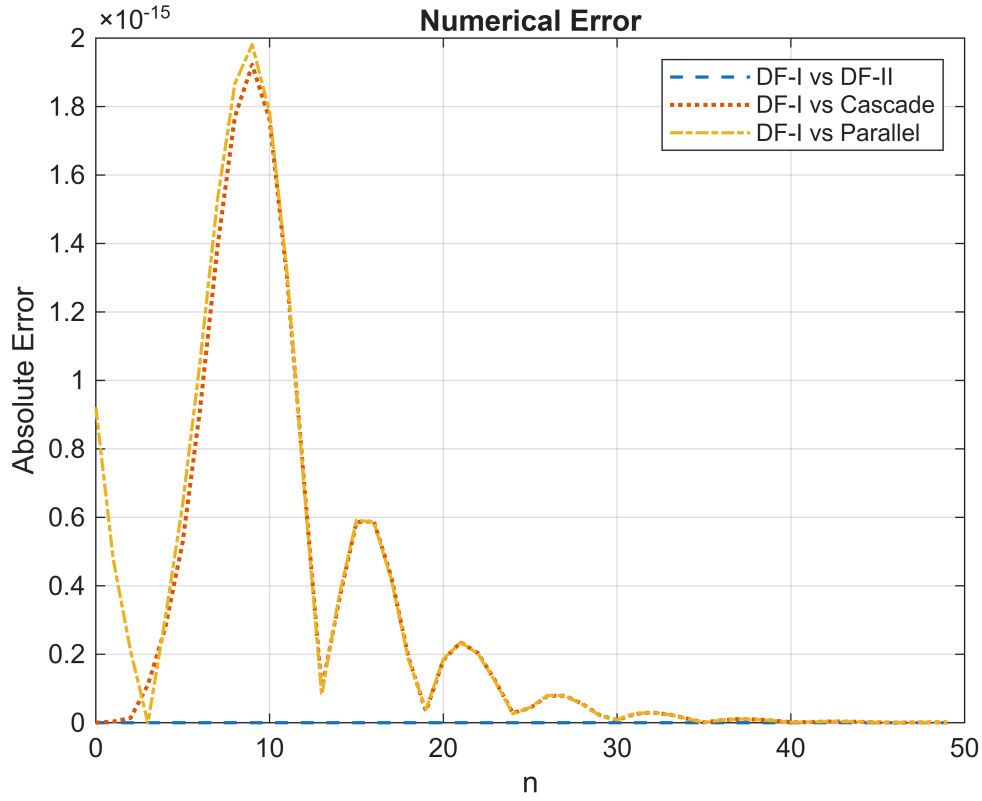



```
%% 7. ERROR PLOT
figure;

plot(n, abs(y1-y2), '--', 'LineWidth', 1.2);
hold on;
plot(n, abs(y1-yc), ':', 'LineWidth', 1.5);
plot(n, abs(y1-yp), '-.', 'LineWidth', 1.2);

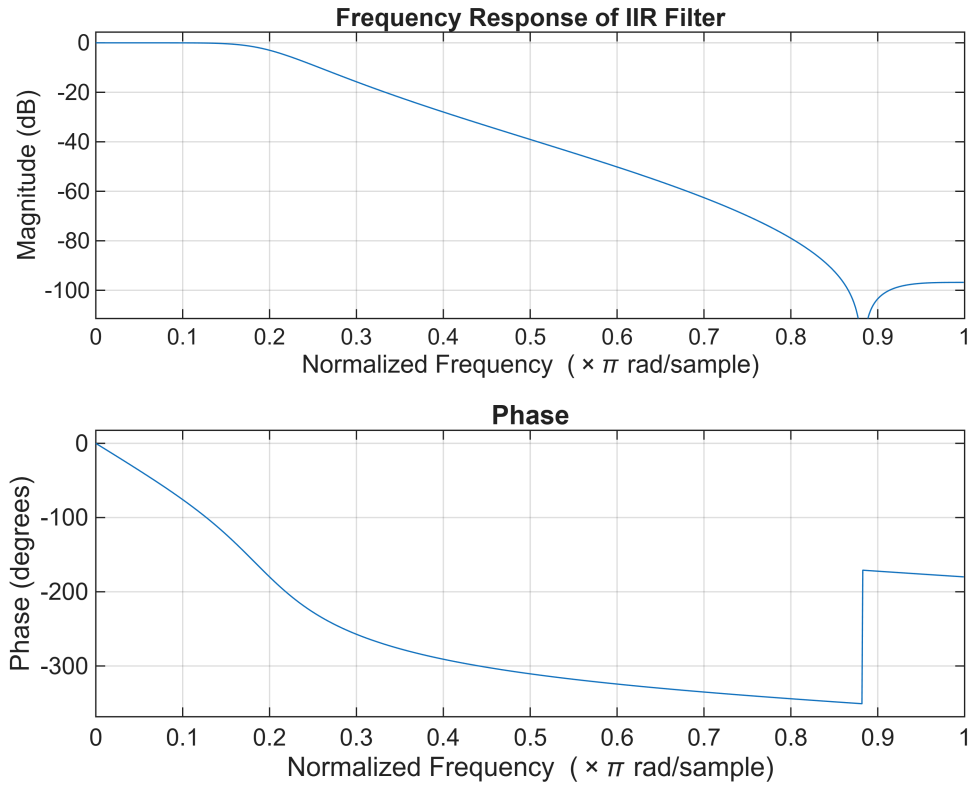
grid on;
xlabel('n');
ylabel('Absolute Error');
title('Numerical Error');

legend('DF-I vs DF-II', ...
       'DF-I vs Cascade', ...
       'DF-I vs Parallel');
```



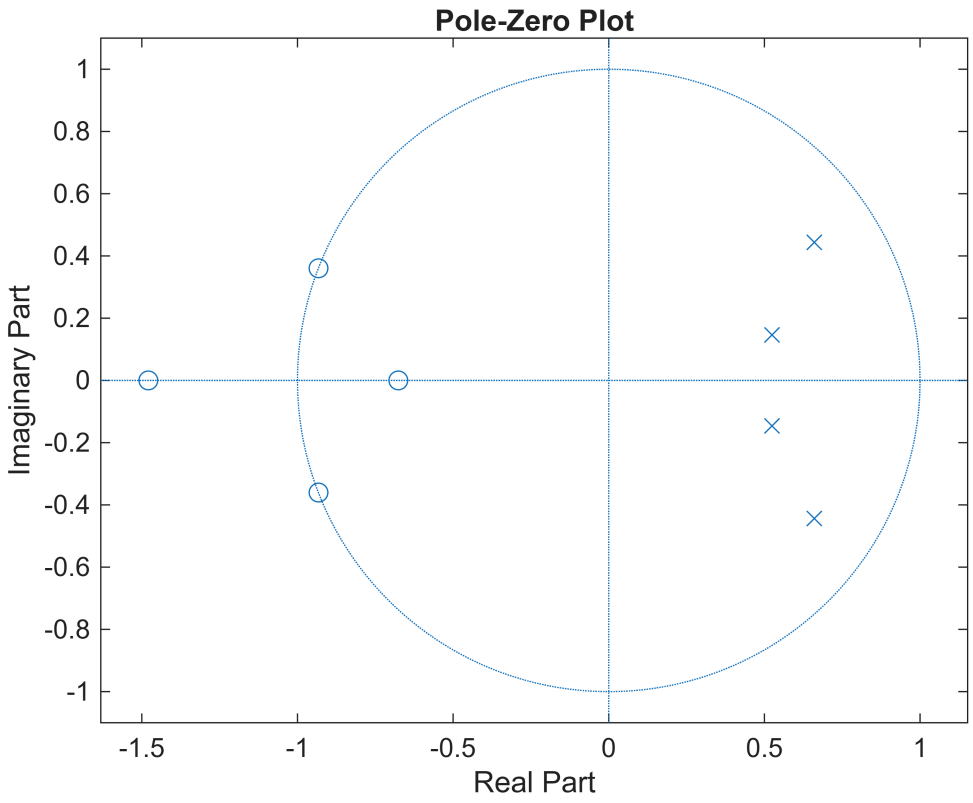

```
%% 8. FREQUENCY RESPONSE
figure;

freqz(b, a, 1024);
title('Frequency Response of IIR Filter');
```




```
%% 9. POLE-ZERO PLOT
figure;

zplane(b, a);
title('Pole-Zero Plot');
```


Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1